Hardware Engineering as a Career

Roles, paths, interviews, portfolio — for the next 10 years

EE • PCB • Firmware • RF • Power • Test

Audience: Final-Year Engineering Students

Roles • Skills • Interviews • Portfolio • Indian market

Why Hardware? Why Now?

Reasons to consider hardware in 2026

AI accelerators

Hardware AI demand is exploding.
NVIDIA, AMD, Google TPUs.

EV revolution

Indian EV market
grew 200% in 2024-25.

IoT everywhere

Wearables, smart homes,
industrial monitoring.

Re-shoring

PLI scheme — semis & PCBs
moving to India.

Less competition

5× fewer hardware engineers
than software in India.

Tangible impact

Build physical things
people use.

Hardware Engineering Roles

Many flavours — pick what excites you

PCB design

Schematic + layout +
manufacturing handoff.

Firmware

C/C++ on MCUs,
drivers, RTOS, comms.

Analog / Mixed-Signal

Op-amps, ADCs, references,
signal conditioning.

Power electronics

SMPS, motor drives,
EV charging.

RF engineer

Radio, antenna, EMC,
compliance.

Test engineer

Functional / production
test jigs & firmware.

FPGA engineer

Verilog/VHDL,

ASIC / VLSI

Chip design — verilog +

Mechanical

Enclosure, thermal,

Skill Matrix

What employers actually look for

Skill	Beginner	Intermediate	Senior
Schematic capture	Arduino-style	Hierarchical, multi-MCU	Complex mixed-signal
PCB layout	2-layer, hand-routed	4-layer, controlled-Z	Multilayer HDI / BGA
Firmware (C)	Blink LED	RTOS, drivers, OTA	Architecture, security
Debug skill	Multimeter	Scope, logic analyzer	Spectrum analyzer, VNA
Power	Linear / buck	SMPS layout & loop comp	SMPS topology design
RF	Use modules	Layout RF section	Antenna design
Manufacturing	Order from JLCPCB	Work with CM, DFM	Set up production line
Soft skills	Document work	Cross-functional review	Lead design teams

Indian Hardware Job Market — 2026

Where the opportunities are

- Bengaluru — biggest hub: ARM, Intel, Qualcomm, AMD, TI, Bosch, Mercedes
- Hyderabad — Samsung, Cyient, Cisco, NetApp, Qualcomm RF
- Pune — Tesla, Honeywell, KPIT, Tata Elxsi, automotive heavy
- Chennai — Ola Electric, Saint Gobain, Wabco, Renault-Nissan
- Delhi-NCR — Garage / startup scene, Foxconn
- Mumbai — Reliance, financial-tech hardware
- Salary entry: ₹6-10 LPA fresher; ₹15-30 LPA senior; ₹50 L+ specialist
- PLI / SPECS scheme — major government push for semis & electronics manufacturing

Top Hardware Employers in India

Where students go after graduation

Semis

Intel, AMD, Qualcomm,
ARM, NVIDIA, Texas Instruments

Wireless

Samsung, Ericsson,
Nokia, Apple, Reliance Jio

Auto

Tesla, Mercedes, Bosch,
KPIT, Tata, Ola Electric

Industrial

Honeywell, ABB,
Schneider, Siemens

Defense

DRDO, BEL, HAL,
Larsen & Toubro Defense

Startups

Ather, ePropelled,
Detect, Tonbo, ClearTrip

Building a Hardware Portfolio

What hiring managers want to see

- 3-5 personal PCB projects on GitHub — schematic + photos + write-up
- Each project: problem, design choices, what you learned, what you'd improve
- Mix: a microcontroller board, a power supply, a sensor system
- Bonus: contribute to OSH project (Arduino, Adafruit, Raspberry Pi)
- Document failures — re-spins teach more than first-attempt success
- Video demo of each working — YouTube link
- Write blog posts / Medium articles about your designs
- Make a portfolio site: link to your GitHub, projects, resume

What to Expect in Interviews

Practical questions you'll face

- Whiteboard: 'design a low-pass filter for X'
- 'Walk me through your most recent PCB project'
- Reading a given schematic — find the bug
- Component selection — pick MOSFET / op-amp / cap for spec
- Calculate: ripple current, dissipation, time constants
- Soldering test (rare but happens)
- Firmware: write a debouncer / state machine in C
- 'How would you debug X if it didn't work?'
- Project deep-dive: every decision you made
- Team dynamics: 'tell me about a conflict you solved'
- Long-form: take-home design problem
- Often: lab tour where they see your hands-on instinct

Internships — The Real Education

Beats any classroom

- Hardware-specific portals: Internshala, AngelList, LinkedIn
- Cold-mail engineering managers — works better than HR portals
- Apply to startups — get exposure to full product cycle
- Apply to MNCs — get exposure to scale + process
- Open-source contributions count as 'internship-equivalent' in modern HR
- Build something that company uses → instant offer
- Stipend: ₹15-50k/month at established companies
- Conversion to full-time: very high if you contribute meaningfully

Soft Skills That Matter

Often more decisive than technical depth

Documentation

Schematic notes,
READMEs, design reviews.

Communication

Explain your design
to non-EEs.

Curiosity

Ask 'why' before
jumping to 'how'.

Resilience

First boards rarely work.
Iterate fast.

Teamwork

Mechanical / firmware /
QA — partner with all.

Time mgmt

Hardware deadlines
are unforgiving.

Specialisation Tracks

After 2-3 years of generalist work

Power electronics

EVs, solar inverters, datacenter PSUs.

High-speed digital

DDR/PCIe/Ethernet, FPGAs.

RF / wireless

5G, sub-GHz, satellite communication.

Mixed-signal / ADC

Audio, ultra-low-power instrumentation.

Automotive

AEC-Q, safety-critical, functional safety.

Embedded ML

TinyML, edge AI, sensor fusion.

Lifelong Learning Resources

Stay sharp after college

- Books: Henry Ott (EMC), Eric Bogatin (SI), Erickson (Power)
- YouTube: Phil's Lab, Robert Feranec, EEVblog, Marco Reps
- Conferences: ESC, Embedded World, India Electronics Week
- MOOCs: edX power-electronics (Erickson), Coursera SI (CU Boulder)
- Coursera 'Building Open-Source Hardware' (free audit)
- Magazines: Circuit Cellar, Elektor, EDN
- Podcasts: Embedded.fm, The Amp Hour, Microcontroller Compendium
- Local meetups: Bangalore Hardware Meetup, ChennaiHWG

Career Mistakes to Avoid

Lessons from senior engineers

- Specialising too early — wait until you've done 2-3 different domains
- Skipping documentation — your future self pays the price
- Refusing to switch tools — Altium snobs and KiCad snobs both lose
- Avoiding firmware — every hardware engineer must read & write C
- Not building anything personal — recruiters need evidence
- Staying in one company > 5 years as a junior — pay grows at switches
- Ignoring soft skills — promotion needs people, not just chips
- Burning out — sustainable pace > heroic pace

Key Takeaways

Key takeaways from this lecture

Hardware is hot

AI, EV, IoT, PLI — demand rising.

Build a portfolio

3-5 personal PCB projects + GitHub.

Internships open doors

Cold-mail, contribute, convert.

Generalist first, then specialise

Try 2-3 domains in 3 years.

Soft skills matter

Communicate, document, teamwork.

Keep learning

Books, YouTube, MOOCs, meetups.

Recommended Resources

For job search & career growth

- LinkedIn India — search by 'PCB Designer / Hardware Engineer'
- Naukri / AngelList — hardware-specific filters
- Embedded.fm podcast — career advice from senior engineers
- The Amp Hour podcast — Chris Gammell hardware career stories
- Local: BHM (Bangalore Hardware Meetup), Chennai HWG, BHQ Mumbai
- Glassdoor / AmbitionBox — salary research
- Cracked the Hardware Interview (book, modest content)
- OSHWA + Open-Hardware project contributions = portfolio gold

Questions & Discussion

Ask anything — concepts, projects, sourcing, career.

References & Further Learning

- LinkedIn India Hardware Engineer search
- Embedded.fm + The Amp Hour podcasts
- Bangalore Hardware Meetup, Chennai HWG
- Embedded World / ESC conference (annual)
- edX 'Embedded Systems' courses
- Coursera 'Building Open Source Hardware'
- Industry mentors via LinkedIn DM
- College alumni networks — your strongest resource

Thank you • Build something this week.